

INTRODUCTION TO MACHINE LEARNING

Learning, Examples, and Data

An overview of the various forms of data and
examples of their use in machine learning.

What are* data?

Anything the world leaves behind that we can **record**.

numbers

sound

images

video

language

games

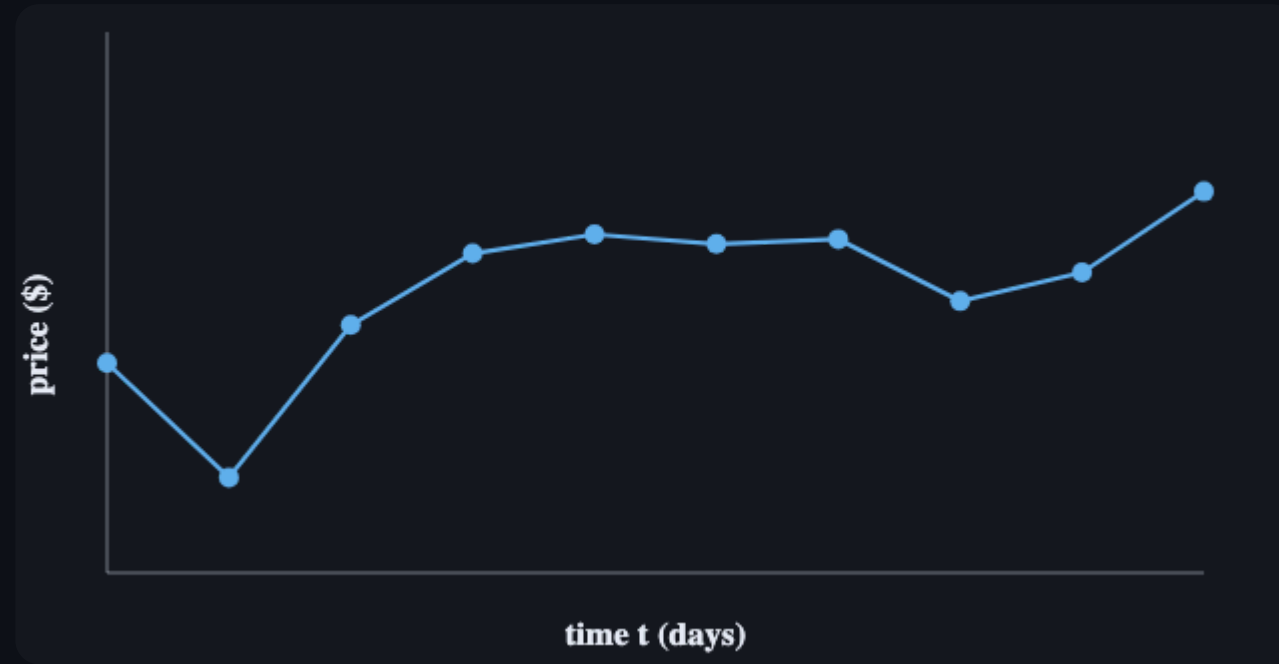
networks

Today: all of these become **numbers**.

Stocks

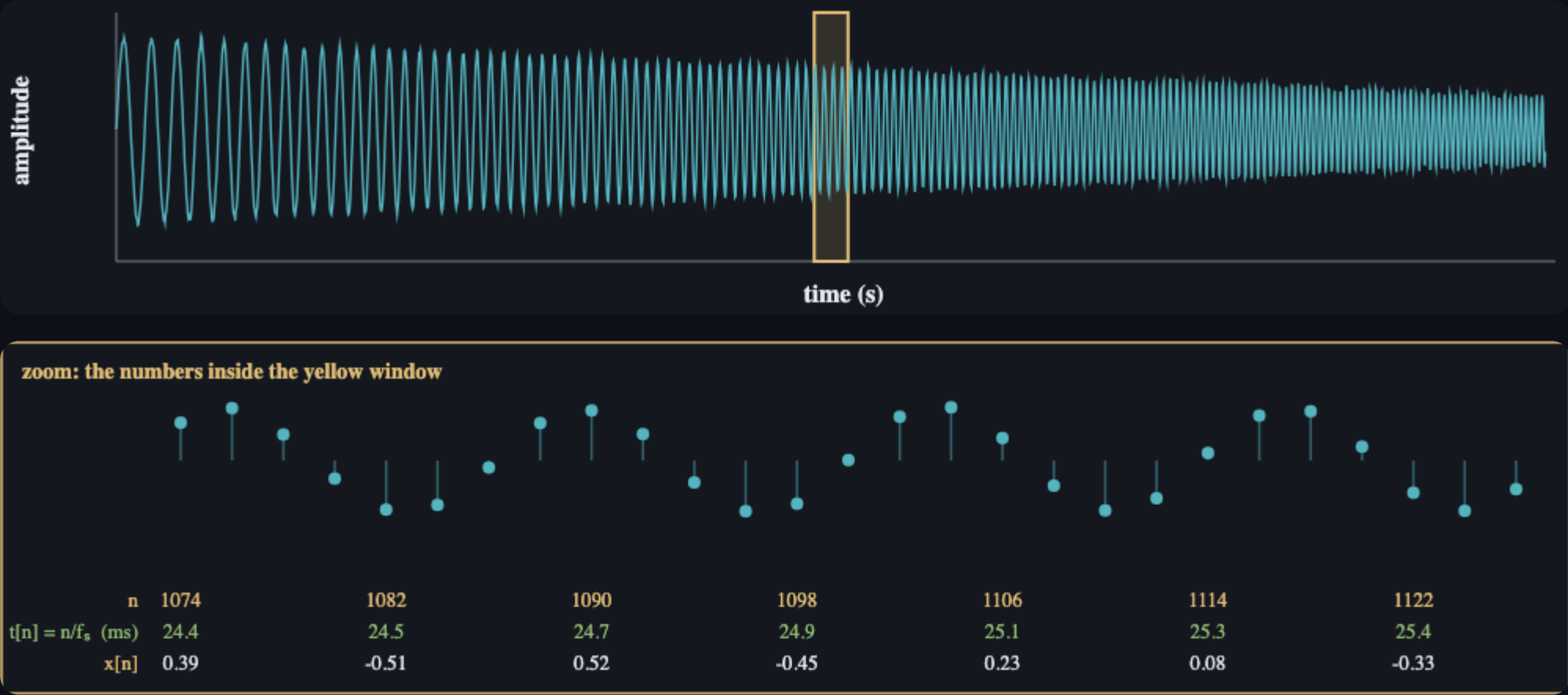
t	price
1	45.4
2	43.0
3	46.2
4	47.7
5	48.1
6	47.9
7	48.0
8	46.7
9	47.3
10	49.0

hover a row



- A **table** and a **curve**: the same object
- Each row is a pair (t_i, x_i)

Audio



play this array

hover the waveform to zoom in · 44,100 numbers per second

- Speech recognition is a function $F : \text{array} \rightarrow \text{text}$

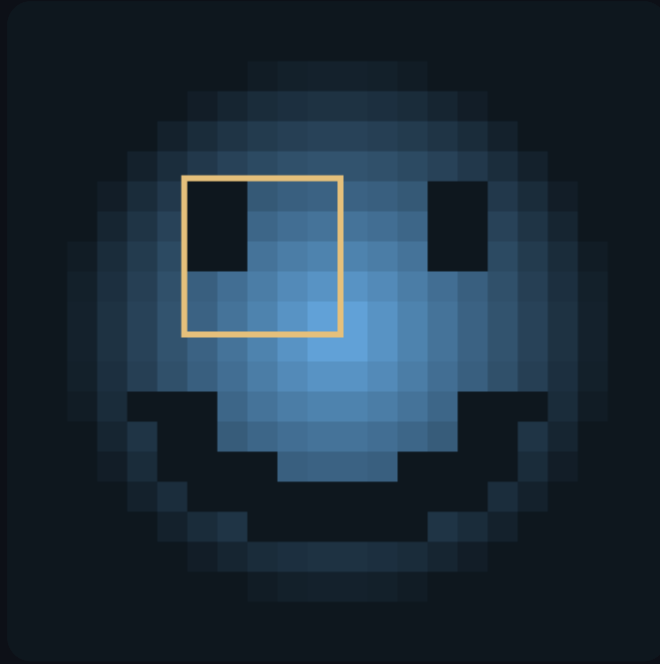
From a Wave to a Picture

The short-time Fourier transform: the first thing anyone does to a sound before modelling it



- The waveform is one number per instant. It hides **which frequencies** are present.
- Cut the signal into short windows, take the Fourier transform of each, stack them side by side. Now pitch is visible.
- Speech recognisers and birdsong classifiers almost never see the wave. They see **this picture**.

Images



30	30	118	127	131
30	30	136	147	153
30	30	153	166	174
127	147	166	184	196
131	153	174	196	215

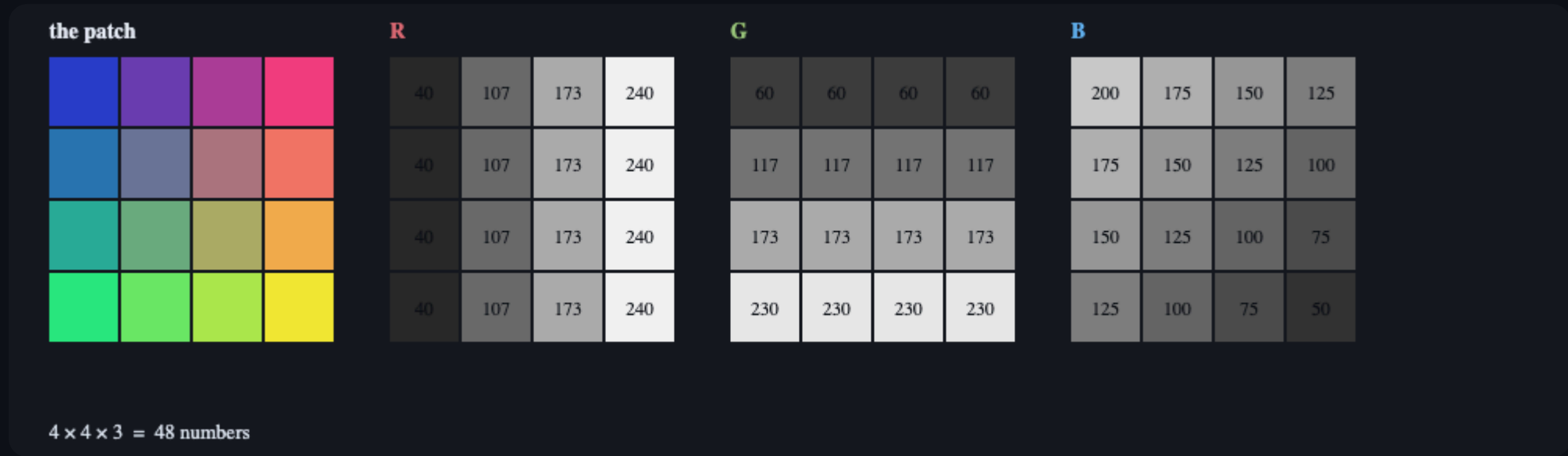
pixel values (blue channel)

move your mouse over the image

- A color image is $h \times w \times 3$: height, width, and **three channels** (red, green, blue)
- Your phone: $3000 \times 4000 \times 3 = 36$ million numbers per photo

Colour Is Three Matrices

The grey image was one number per pixel. Colour is three, stacked



all three

red only

green only

blue only

a 4 by 4 patch, and the numbers behind it

- A colour image of height h and width w is an array of shape $h \times w \times 3$.
- Nothing about it is a picture to the machine. It is three grids of numbers between 0 and 255.

Video

Eadweard Muybridge, 1878: the first time motion was recorded as an array of stills



play the stack

click any frame to move the square · the grid is the real pixels under it

t = 1 of 12

- One frame is $h \times w \times 3$; a clip is $T \times h \times w \times 3$
- One minute of 1080p at 30 fps: about **11 billion numbers**
- Muybridge settled a bet: are all four hooves ever off the ground at once? You cannot see it, so he **turned it into data**.

Frames from [The Horse in Motion](#), Wikimedia Commons, public domain.

Image Recognition

- **2012**: Google Brain finds cats in YouTube, unsupervised
- **2014**: face recognition reaches 97%
- **2019**: deep models match radiologists on lung scans
- **2021**: AlphaFold predicts protein structure from sequence
- **2022**: text becomes images, and images become text
- **2024**: one model reads an image, a page and a recording together
- **2026**: two lines of prompt produce fifteen seconds of film

The same training recipe each time, on **new data**.



2019: lung cancer detection from chest scans

[Ardila et al., Nature Medicine 2019](#)

Text

One-hot word representation

	The	World	Is	A	Crazy	Place
dictionary	the	world	is	a	crazy	place
a	0	0	0	1	0	0
crazy	0	0	0	0	1	0
is	0	0	1	0	0	0
learning	0	0	0	0	0	0
machine	0	0	0	0	0	0
place	0	0	0	0	0	1
the	1	0	0	0	0	0
world	0	1	0	0	0	0

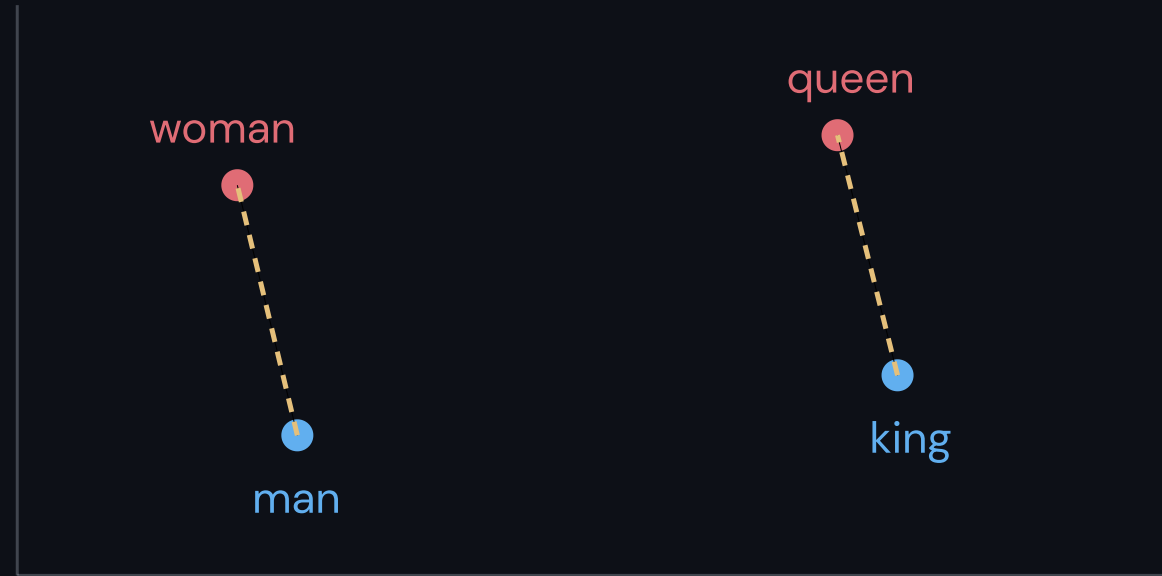
one column per word: a 1 in its dictionary slot

hover a word: its column lights up

- **One-hot** is honest but lonely: every word equally far from every other

Text

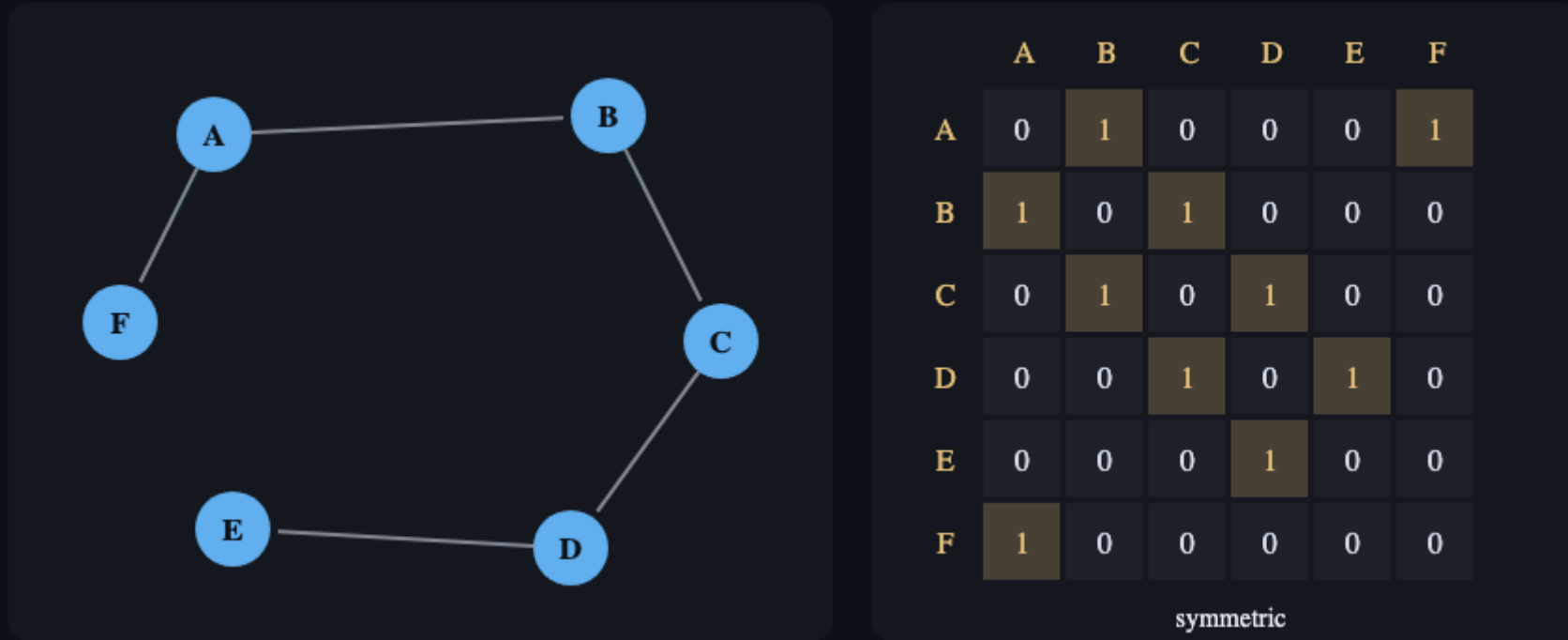
Word embedding representation



DIRECTION MEANS SOMETHING

$$\vec{v}_{\text{king}} - \vec{v}_{\text{man}} + \vec{v}_{\text{woman}} \approx \vec{v}_{\text{queen}}$$

Graph Representation



undirected

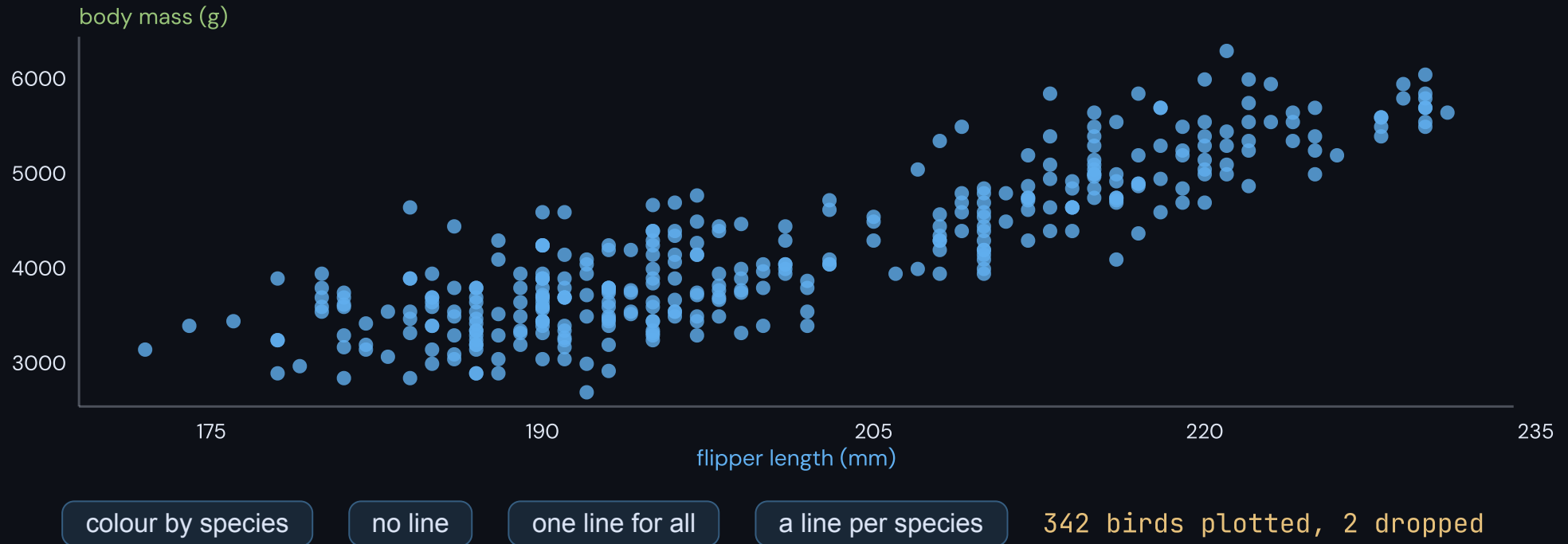
directed

click two nodes: the matrix follows

- Maps, molecules, social networks: all **matrices**, $A_{ij} = 1$ when i connects to j

A Real Dataset

344 penguins, measured at Palmer Station, Antarctica. Nobody made these numbers up.



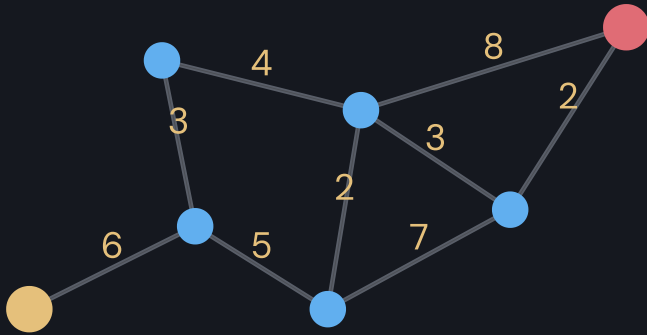
- 11 of 344 rows are incomplete; 2 are missing what *this* plot needs. Which holes matter depends on the question.
- One line for everything says 50 g per mm. Colour by species and it starts to look like an answer to a question nobody asked.

Data: [Palmer Penguins](#), collected by Dr Kristen Gorman, Palmer Station Antarctica LTER. CCO. [Download the CSV](#)

The Three Pillars of Artificial Intelligence

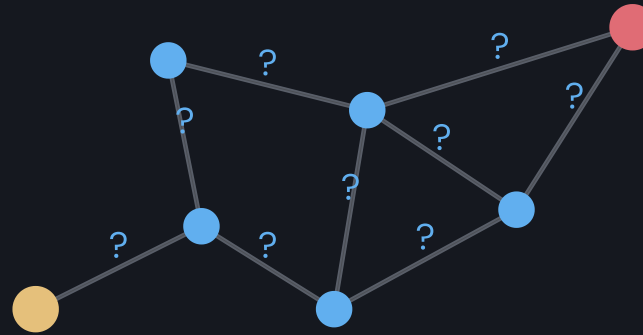
The world is messy. The question is simple: **what is the shortest way home?**

MODELING



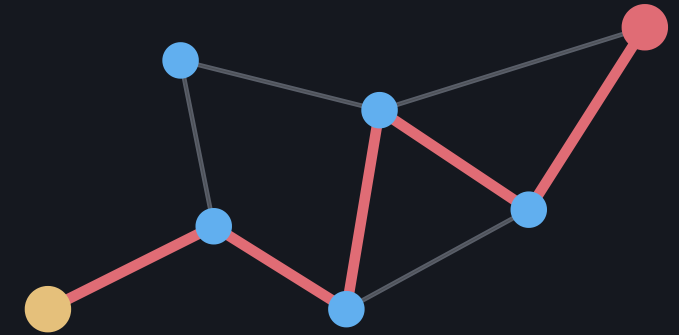
Throw the city away. Keep junctions and the cost of each road.

LEARNING



You are not given the costs. You estimate them from journeys people already made.

INFERENCE



cost 18

With a model in hand, answer the question you actually asked.

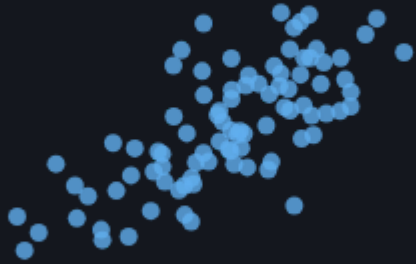
Classification, Regression, Structured Prediction

x	$\xrightarrow{f}$	y	
credit card transaction	→	fraud / not fraud	classification
measurements of a collision event	→	Higgs decay / background	classification
satellite image of a region	→	poverty index	regression
information about a house	→	price	regression
English sentence	→	Japanese sentence	structured
image	→	sentence describing it	structured

Same skeleton every time. Only the shape of y changes.

Now you: think of one input–output pair from your own life. What is x , what is y , and which of the three is it?

Correlation Coefficient (2 Variables)



$$r_{XY} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$$

target  $r = 0.72$

- **Linear** co-movement, from -1 to $+1$
- $r = 0$ means not linear, **not** unrelated

Histogram and Probability Density Function



KEY DISTINCTION

The histogram is **data**. The curve is a **model**.

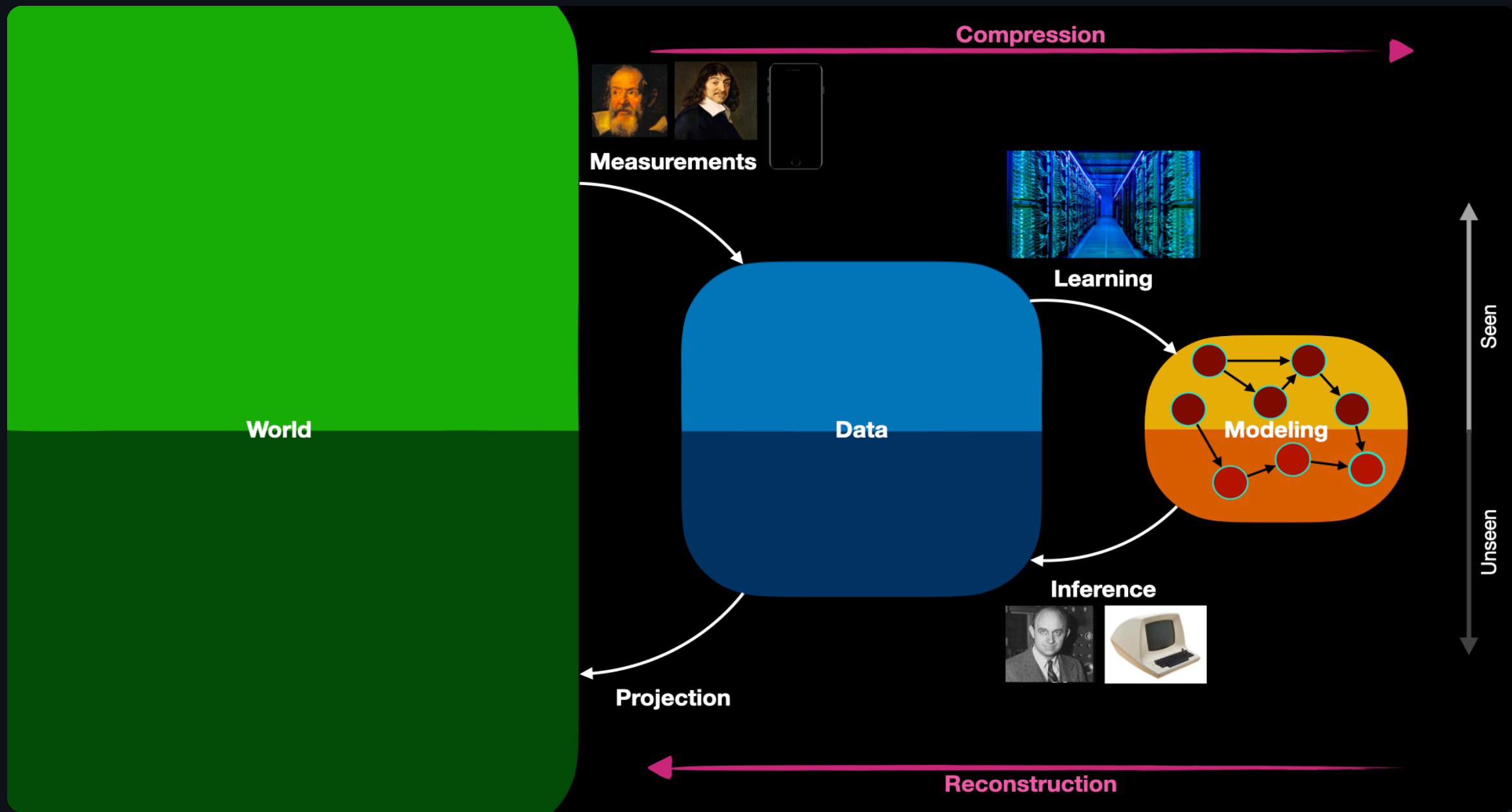
Where to Look for Data?

- **Curated:** Kaggle, UCI, Hugging Face, data.gov
- **Collected:** sensors, experiments, your phone
- **Scraped:** the web, politely

```
import pandas as pd

url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/penguins.csv"
df = pd.read_csv(url)

print(df.shape)      # (344, 7): 344 penguins, 7 measurements
print(df.describe()) # your first look at any dataset
```



NEXT SESSION

Data is the interface between the world and the model.

Thursday: the math and Python toolkit.
Before then: PSO, and run the penguins line yourself.